E-commerce Business Analysis

Driving Revenue Growth and Identify Potential Development

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1. Project Objectives

The objective of this project is to analyse the company's eCommerce performance in the UK market from Jan 2010 to Jun 2011 and generate data-driven insights to support strategic business decision-making. The analysis focuses on evaluating key business areas, including sales performance, customer behaviour, marketing effectiveness and product performance.

Specifically, this project aims to:

1. Evaluate revenue trends over time to determine whether the company's sales performance is improving or declining and identify key factors influencing revenue changes.
2. Identify the best-performing products and categories based on sales volume, revenue contribution, and profitability to understand drivers of business growth.
3. Analyse customer value and behaviour to identify high-value customer segments and assess factors influencing customer retention and long-term profitability.
4. Evaluate the effectiveness of different marketing channels by analysing their contribution to customer acquisition, revenue generation, and customer engagement.
5. Measure customer churn rate and examine whether churn is increasing over time, providing insights into potential retention challenges and improvement opportunities.
6. Assess the impact of promotional activities on business performance by analysing whether discounts and campaigns contribute to increased sales or negatively affect profit margins.
7. Investigate the impact of product returns on overall business performance by examining their effects on revenue, profitability, and production costs, and identifying potential areas for improving operational efficiency.

Through this analysis, the project aims to provide actionable insights that help the company optimise sales strategies, improve customer retention, enhance marketing effectiveness, and increase overall profitability.

2. Data Sources & Transformations

The dataset used in this project was sourced from Kaggle and represents a simulated eCommerce transaction dataset for analytical practice purposes. The dataset contains transaction records from multiple countries; however, this project focuses exclusively on the United Kingdom (UK) market, as approximately 80% of the available data originates from this region. Due to the uneven geographical distribution of records, the dataset was considered unsuitable for conducting reliable cross-market comparisons.

Before analysis, several data cleaning and transformation steps were performed to improve data quality and ensure the dataset was suitable for business reporting. The data types of key columns were standardised, including transaction dates, customer information, numerical measures, and categorical variables. Date fields were transformed into appropriate formats to support time-based analysis.

The dataset contained missing values, primarily within the **Customer ID** column. These missing values are likely caused by customers completing purchases without registering or logging into an account, resulting in unavailable customer-level information. Although these records cannot be used for certain customer behaviour analyses, they were retained because they still contain valid transaction information and provide valuable insights for analysing overall sales performance, revenue, and operational metrics. Removing these records would result in the loss of meaningful transaction data.

To improve data modelling efficiency and support accurate reporting, the dataset was transformed into a star schema structure consisting of one **Fact_Sales** table and multiple Dimension tables, including **Dim_Customer, Dim_Payment and Dim_Product.** A dedicated **Dim_Date table** was also created in Power BI to enable consistent time intelligence analysis, such as revenue trends, seasonal patterns, and year-over-year performance evaluation.

3. Data Model & DAX

The data model was structured using a star schema, where dimension tables were connected to the Fact Sales table through one-to-many relationships. Dimension tables contain descriptive attributes used for filtering and segmentation, while the fact table stores transactional data used for performance measurement.

Several DAX measures were created in Power BI to support the calculation of key performance indicators (KPIs) and provide deeper insights into business performance. While multiple measures were developed throughout the analysis process, only the most significant measures related to the project objectives are highlighted.

The key DAX measures include:

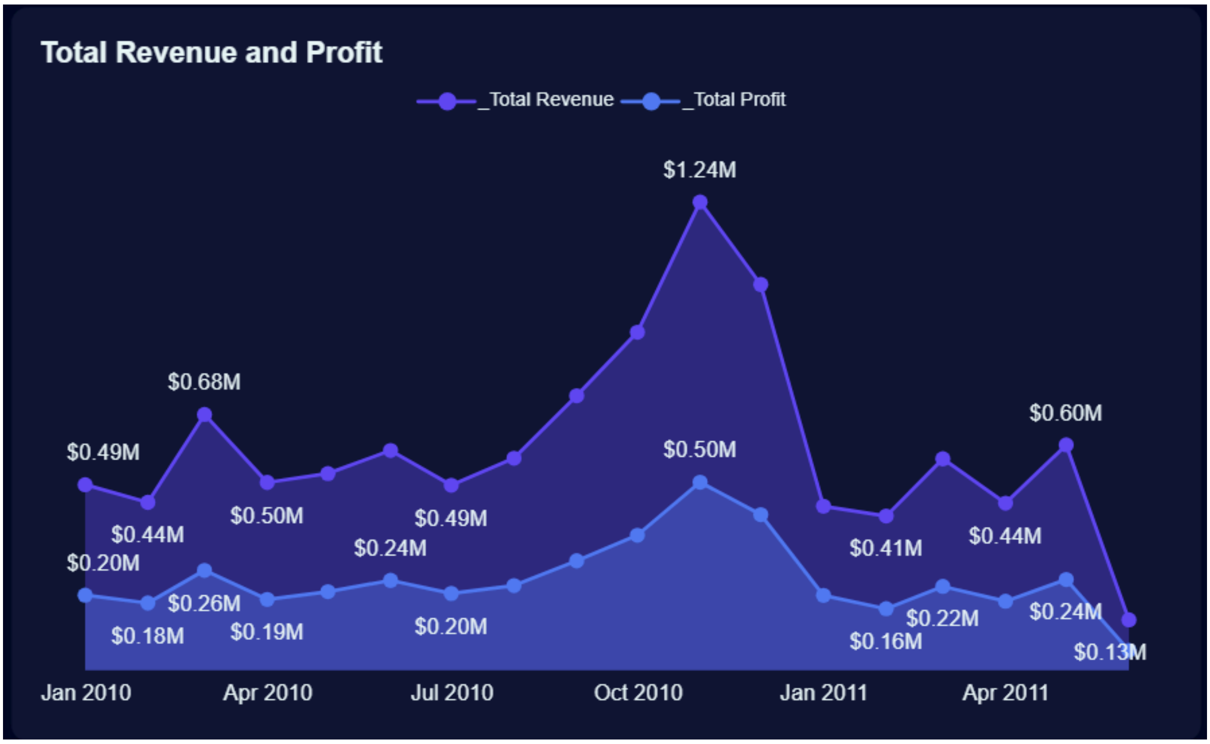
- **Customer Churn Rate:** Calculated the proportion of customers classified as churned to evaluate customer retention performance and identify potential issues affecting customer loyalty.
- **Profit Margin:** Measured business profitability by analysing the percentage of profit generated from revenue, allowing evaluation of product and overall business performance.
- **Customer Age Group Classification:** Created customer age categories to support demographic analysis and identify purchasing patterns across different customer segments.
- **Product Return Rate:** Calculated the percentage of returned products to evaluate return behaviour and assess potential impacts on business operations.
- **Return Product Cost:** Estimated the production cost associated with returned products to analyse the financial impact of product returns on profitability.

These measures were integrated into Power BI visualisations to support analysis of customer behaviour, profitability, operational efficiency, and overall business performance. Additional DAX calculations were developed as supporting measures for specific visualisations but are not discussed in detail as they do not directly contribute to the main analytical objectives.

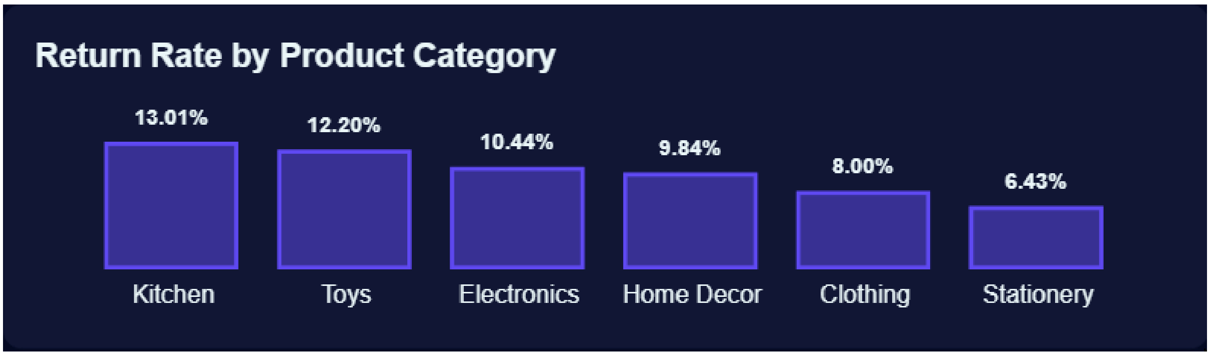
4. **Visuals & Key Insights:** Capture images and describe, summarize the key insights, and provide recommendations if applicable.

Overall Performance

The business generated **$10.72 million** in total revenue and **$4.32 million** in total profit, resulting in a **profit margin of 40.29%** across **34,000 orders**. While the business demonstrates strong profitability, the **customer churn rate of 63.78%** is relatively high, indicating a significant customer retention challenge. In addition, the **return rate of 10.03%** suggests that nearly one in ten purchased products was returned.



Revenue and profit followed a similar trend over the analysis period. Both metrics increased steadily throughout **2010**, reaching their highest levels in **November 2010**, with revenue peaking at approximately **$1.24 million** and profit at around **$0.50 million**. However, both declined sharply after the peak and remained relatively lower during the first half of **2011**.



Product returns varied across categories. **Kitchen** recorded the highest return rate (**13.01%**), followed by **Toys (12.20%)** and **Electronics (10.44%)**, while **Stationery** had the lowest return rate (**6.43%**).

These findings suggest that the Clothing, Toys and Electronics categories may require further investigation into product quality, sizing, or customer expectations.

The monthly sales trend shows that the number of products sold generally exceeded the number of products returned throughout the period. Sales volume peaked at approximately **0.63 million units** in **December 2010**. Although there are still some product returns each month, the number is insignificant compared to the number of products sold.

Key Insights

1. **Strong profitability provides a solid foundation for growth:** The business generated **$10.72M in revenue and $4.32M in profit**, with a **40.29% profit margin**, indicating strong overall financial performance and room to invest in customer retention and operational improvements.

2. **Customer retention is the most significant strategic concern:** A **63.78% churn rate** suggests that a large proportion of customers do not make repeat purchases. This represents a major opportunity to improve **customer lifetime value (CLV)** through retention campaigns, personalised offers, and targeted re-engagement.

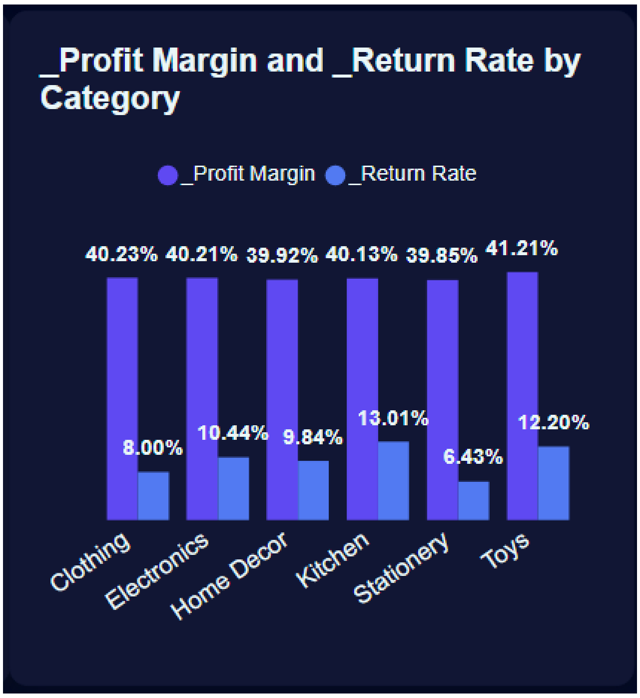
3. **Product returns create a meaningful profitability risk:** With a **10.03% return rate**, returns may be contributing to unnecessary costs and lost revenue. The issue should be investigated at the **category and product level** to identify potential drivers such as product quality, inaccurate descriptions, or customer expectations.

4. **Sales performance is highly seasonal:** Revenue, profit, and sales volume peaked in **March, May and November**, highlighting a strong seasonal demand pattern. The business should leverage this period through **inventory planning, promotional campaigns, and marketing investment** to maximise peak-season performance.

5. **Kitchen, Toys, and Electronics require return-rate investigation:** These categories recorded the highest returns, making them priority areas for further analysis. Identifying the

specific products and subcategories driving returns could help reduce avoidable costs and protect profit margins.

Sales Performance



The business demonstrates strong product profitability, with an overall profit margin of approximately 40%. Profit margins remain relatively consistent across product categories, with Toys, Stationery, and Clothing achieving margins above 40%. Home Decor records a slightly lower margin at 39.9%, but it remains highly profitable compared with other categories.

Kitchen, Toys and Electronic have the highest return rate, accounting for higher than the overall return rate (10.03%).

Toys, Stationery, Clothing, and Home Decor are the strongest-performing categories, generating the highest sales volume (more than 1 million units sold) and making significant contributions to overall revenue. These categories represent the core revenue drivers of the business, indicating strong customer demand. Business should remain a focus area for inventory management and marketing strategies for these product categories.

However, it is important to note that the dataset only covers the period from January 2010 to June 2011; therefore, the analysis does not represent the complete annual sales cycle. Despite this limitation, the monthly order pattern shows consistent seasonal trends, with peak periods occurring in March 2010, May 2010, and November 2010. November recorded the highest number of orders, exceeding 3,000 orders, while March and May achieved more than 2,000 orders. A similar pattern was observed in 2011, where March and May were also among the highest-performing months. This suggests that customer purchasing behaviour may follow recurring seasonal patterns.

Key Insights

1. **Core categories drive overall performance**: **Toys, Stationery, Clothing, and Home Decor** are the key volume-driving categories, each exceeding **1 million units sold** and making a significant contribution to total revenue. These categories should remain a **strategic focus for inventory planning, marketing investment, and revenue growth**.

2. **Seasonality presents an opportunity for targeted planning:** Despite the dataset covering only **January 2010 to June 2011**, recurring peaks were observed in **March, May, and November**, with **November 2010 exceeding 3,000 orders** and representing the strongest observed month. This suggests an opportunity to align **inventory, staffing, and promotional activity** with periods of higher customer demand.

3. **Returns represent a potential profitability and operational risk: Toys generated the highest sales volume and revenue while also recording a high number of returns**, indicating that strong sales performance does not necessarily translate into optimal operational efficiency. **Kitchen**, despite having lower sales volume, recorded the **highest return rate**, making it a priority for further investigation into product quality, customer expectations, or product-specific issues.

4. **Optimising high-performing categories can unlock further value:** The combination of **high sales volumes and healthy margins** demonstrates strong performance across the leading categories. However, analysing **return patterns, product-level performance, and customer purchasing behaviour** within these categories could identify opportunities to reduce avoidable costs and further improve profitability.

Cost Performance

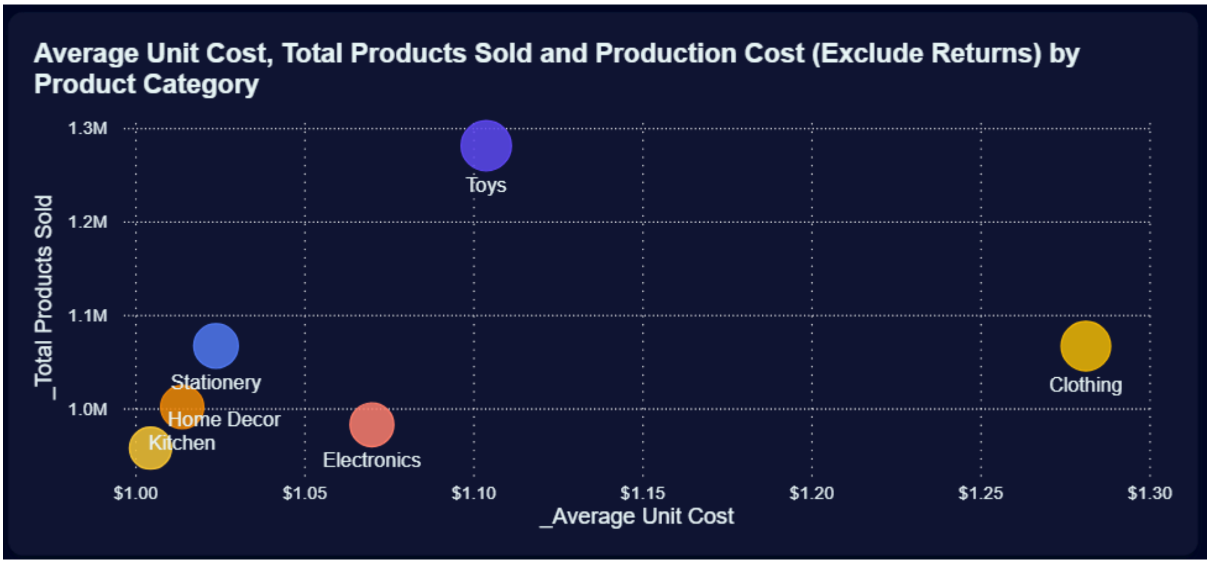
Category	age Unit Cost	_Total Products Sold	_Total Production Cost (Exclude Returns)	_Profit Margin
⊞ Toys	$1.10	1,280,788	1,413,833.94	41.21%
⊞ Clothing	$1.28	1,066,347	1,366,272.52	40.23%
⊞ Stationery	$1.02	1,066,546	1,092,076.85	39.85%
⊞ Electronics	$1.07	982,110	1,050,919.69	40.21%
⊞ Kitchen	$1.01	1,001,116	1,015,074.56	40.13%
⊞ Home Decor	$1.00	957,236	961,610.20	39.92%
Total	$1.09	6,354,143	6,899,787.76	40.29%

Product categories performance

Toys: Toys generate the largest absolute return volume and the highest return-related production cost. The high return cost is partly associated with the category's large sales volume, but its 12.20% return rate also indicates that returns remain significant relative to sales.

Kitchen: Kitchen has the highest return intensity relative to its sales volume. Although Kitchen does not generate the highest return-related cost, it has the highest proportion of sold units being returned. -> **require further investigation at product/subcategory level**

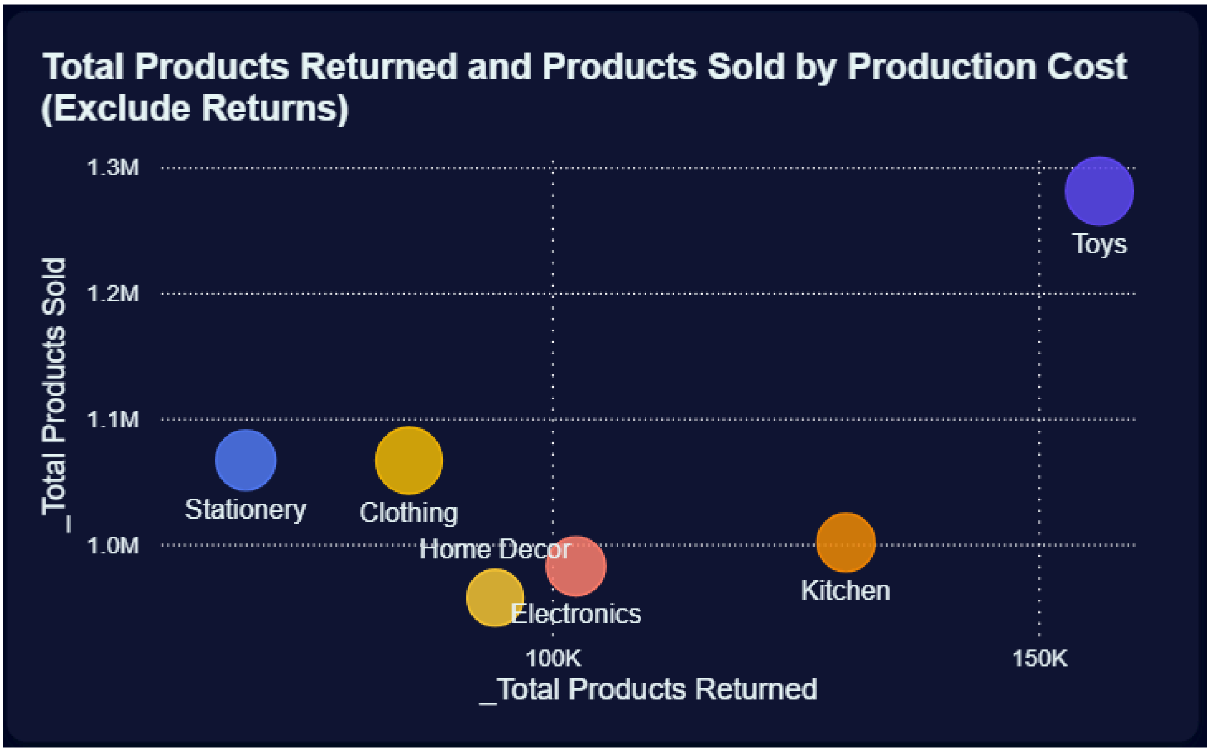
Clothing: has lower return rate compared to the overall return rate; however, it stands for the second highest total production cost. **Clothing has relatively low return volume but disproportionately high return production cost.**



The analysis highlights significant differences in sales contribution, revenue generation, cost efficiency, and return behaviour across product categories. Toys and Stationery are the strongest-performing categories in terms of sales volume, with both categories selling more than 1 million units and contributing substantially to overall revenue. However, sales volume alone does not fully explain revenue performance.

Clothing demonstrates a stronger revenue contribution despite having a lower sales volume compared with Toys and Stationery. Although Clothing sold approximately 300K fewer units than Toys and 100K fewer units than Stationery, it generated a comparable level of revenue. This suggests that Clothing products have a higher average selling value and may contribute greater revenue per unit compared with other high-volume categories.

Several categories, including Kitchen, Home Decor, and Electronics, generated strong revenue performance despite having relatively lower sales volumes (below 1.1 million units). These categories generated approximately $1.8 million in revenue, indicating higher-value products or stronger pricing potential. This highlights opportunities for the business to focus not only on sales volume but also on product value and revenue contribution.



From a cost perspective, Toys and Clothing have the highest production costs. Notably, Clothing incurred the highest total production cost despite selling fewer units than Toys, suggesting that Clothing products may require higher production investment or have higher unit costs. Further analysis of cost per unit would help identify whether the higher cost is driven by product complexity, material costs, or pricing strategy.

Return analysis reveals that Toys have the highest number of returned products, which is expected given their high sales volume. However, Kitchen requires further investigation as it records the second-highest number of returns despite having a lower sales volume. This may indicate potential issues related to product quality, customer expectations, product descriptions, or purchasing behaviour.

Overall, Stationery appears to be the most balanced-performing category, achieving high sales volume, strong revenue contribution, relatively low production costs, and the lowest return volume among categories. This indicates that Stationery represents a potential growth opportunity due to its combination of profitability, operational efficiency, and customer satisfaction.

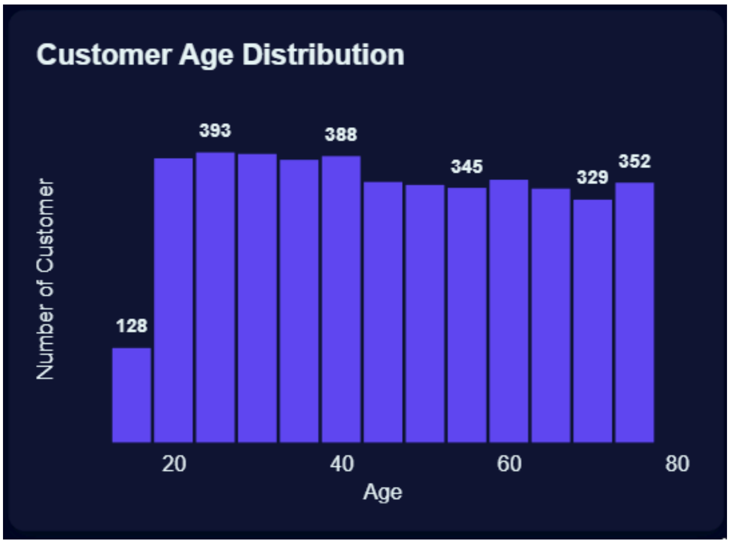
Although returned products represent a smaller proportion compared with total products sold, they still create financial impacts because production costs have already been incurred. During the analysed period, returned products represented more than $500K in production costs, highlighting the importance of reducing avoidable returns to protect profitability.

Key Insights

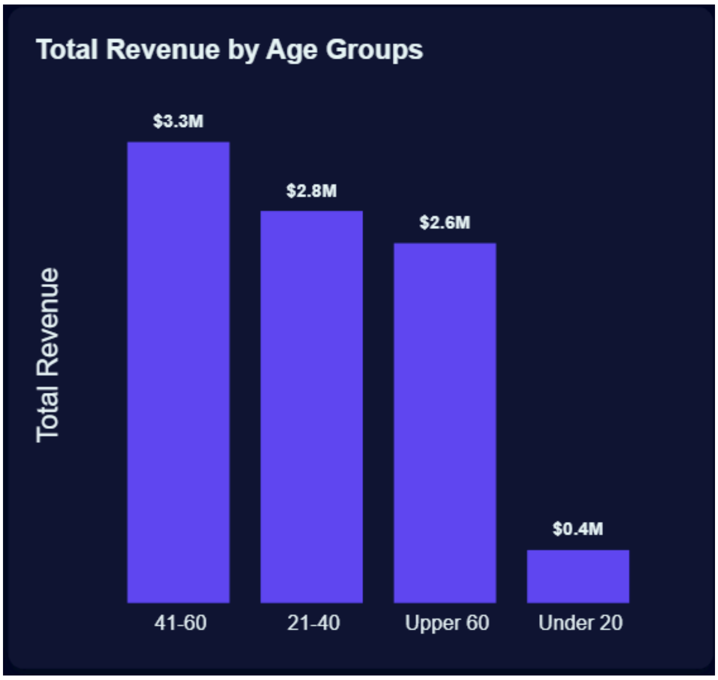
1. **Strong category performance:** Toys, Clothings and Stationery are the highest-volume categories, each exceeding **1 million units sold** and making significant contributions to overall revenue.
2. **High-value category opportunity:** Clothing generates revenue comparable to Toys and Stationery despite selling fewer units, suggesting higher average product value and stronger revenue contribution per unit.
3. **Cost efficiency concern:** Clothing and Toys have the highest production costs. Clothing requires further investigation as it has the highest total production cost despite lower sales volume than Toys.
4. **Return management opportunity:** Returned products resulted in more than **$500K in production costs**, representing a potential profitability improvement area despite returns accounting for a smaller proportion of total sales.
5. **Kitchen category risk:** Kitchen products have relatively lower sales volume but record the highest number of returns, indicating potential issues with product quality, customer expectations, or product information.
6. **Stationery as a growth opportunity:** Stationery demonstrates a strong balance between sales volume, revenue generation, low production cost, and low returns, making it a potential category for further investment.

Customer Segment

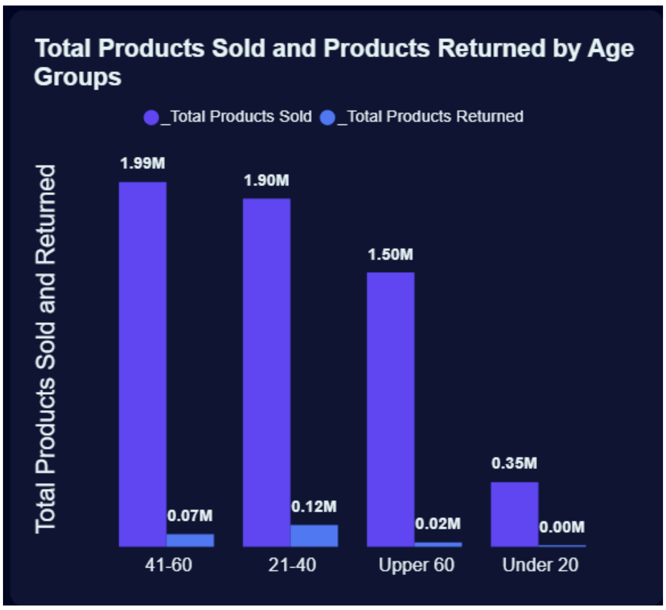
The customer base is relatively balanced across gender, with females accounting for approximately 50.8% of customers and males 49.2%. This indicates that the business attracts both genders equally, suggesting that marketing campaigns and product offerings have broad customer appeal rather than targeting a single gender segment.



The customer population is concentrated within the 21–60 age range, with the 21–40 age group representing the largest customer segment, followed closely by customers aged 41–60. In contrast, customers under 20 represent only a small proportion of the customer base, indicating that the business primarily serves adult consumers with stronger purchasing power.



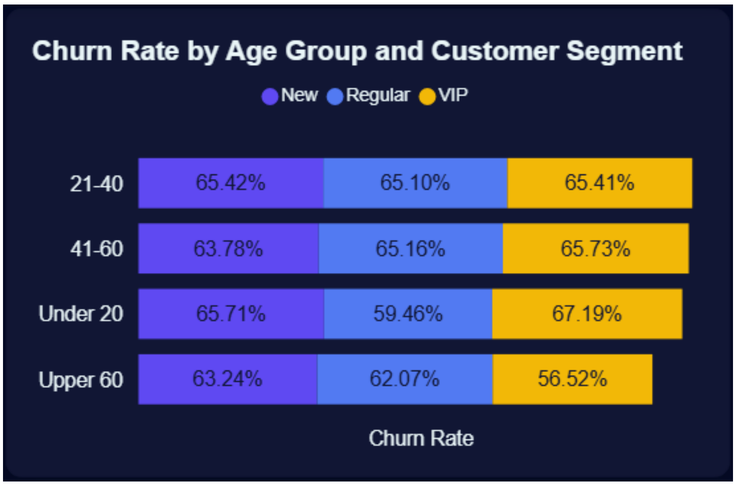
Customer purchasing behaviour aligns closely with the customer distribution. Customers aged 41–60 generated the highest revenue (approximately $3.3M) and purchased around 2 million products during the analysis period. The 21–40 age group ranked second, generating approximately $2.8M in revenue with 1.9 million products sold. Together, these two age groups represent the primary revenue-driving customer segments and should remain the focus of customer acquisition and retention strategies.



Despite generating the highest revenue, the 41–60 age group also recorded the largest number of returned products. However, this appears to be largely driven by higher purchasing volume rather than an unusually high return tendency. Analysing return rates by age group would provide a more accurate assessment of customer satisfaction and purchasing behaviour.

Despite ranking third in total product volume, the **60+ age group records one of**

the lowest return volumes, comparable to other high-performing age groups. This indicates a **strong revenue opportunity**, as the segment combines substantial purchase volume with relatively low product returns, making it a potentially valuable customer segment for **targeted retention and marketing strategies**.



Customer retention represents the most significant business concern. The overall churn rate is approximately 63.8%, indicating that nearly two-thirds of customers did not make repeat purchases during the observed period. Churn remains consistently high across all age groups and customer segments, generally ranging between 60% and 67%. This suggests that customer attrition is a business-wide issue rather than one isolated to a specific demographic.

Although VIP customers are typically expected to demonstrate stronger loyalty, the analysis shows that churn rates remain relatively high across all customer segments. This may indicate opportunities to strengthen customer engagement through personalised promotions, loyalty programs, and post-purchase communication to improve long-term customer retention.

Key Insights

1. **Customer retention is the biggest challenge:** The overall **churn rate is 63.78%**, indicating that a large proportion of customers do not return for future purchases.

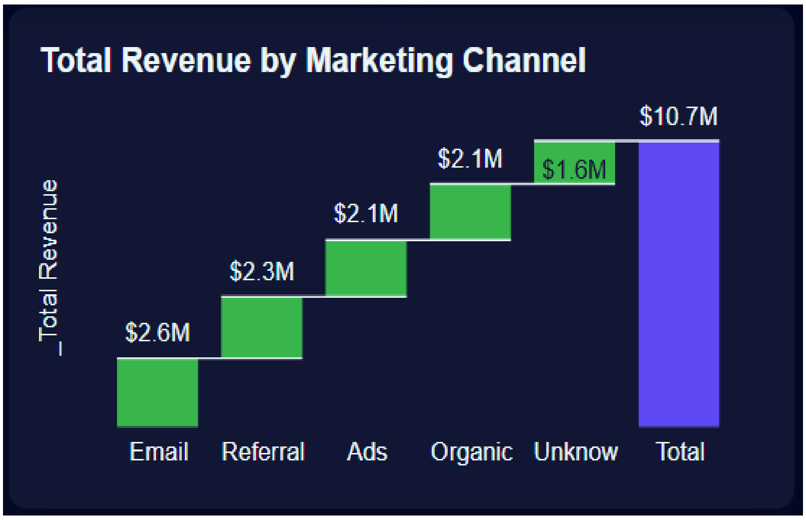
2. **Revenue is driven by adult customers:** Customers aged **21–60** contribute the majority of sales, with the **41–60** segment generating the highest revenue (**~$3.3M**) and sales volume (**~2 million products**).

3. **Balanced customer demographics:** The customer base is almost evenly split between **female (50.8%)** and **male (49.2%)** customers, suggesting broad market appeal across genders.

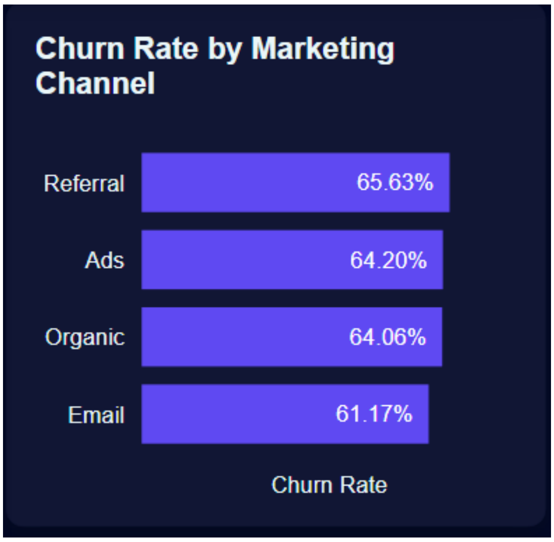
4. **Consistently high churn across segments:** Churn remains above **60%** across all age groups and customer segments, indicating that customer retention should be prioritised across the entire customer base rather than targeting a single demographic.

5. **Returns follow purchasing volume:** Customers aged **41–60** record the highest number of returned products, largely reflecting their higher purchasing volume rather than necessarily poorer customer experience.

6. **Opportunity to improve customer loyalty:** Strengthening loyalty programs, personalised marketing campaigns, and post-purchase engagement may help reduce churn and increase customer lifetime value.

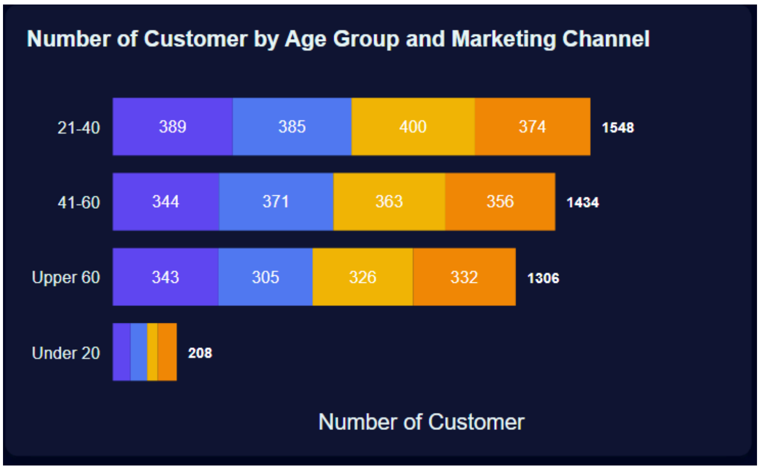


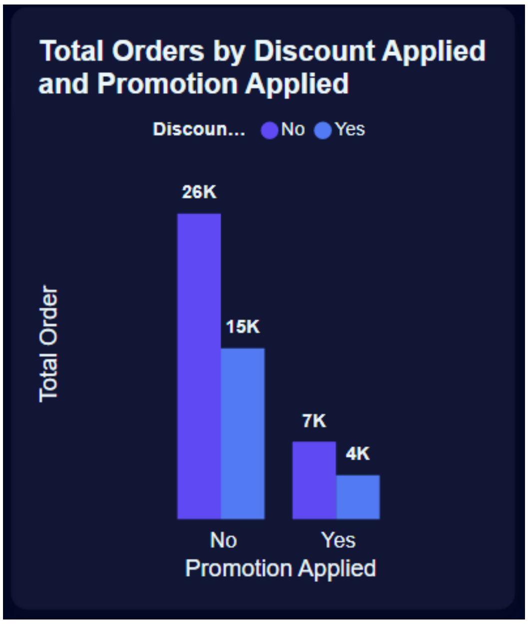
Marketing performance is relatively balanced across acquisition channels, with Email generating the highest revenue (approximately $2.6M) and profit (approximately $1.0M). Referral, Ads, and Organic channels also contribute comparable levels of revenue, indicating that the business is not overly dependent on a single marketing channel for growth. This diversified channel performance reduces acquisition risk and demonstrates a well-distributed customer acquisition strategy.



Customer retention remains a challenge across all marketing channels. Churn rates range from approximately 61% to 66%, with Referral recording the highest churn rate (65.63%) and Email the lowest (61.17%). Although the differences are relatively small, Email appears to acquire customers with slightly stronger long-term retention, while Referral campaigns may require further optimisation to improve customer loyalty.

Customer acquisition by marketing channels is also relatively consistent across age groups. The 21–60 age groups are well represented across Email, Organic, Ads, and Referral, suggesting that no single channel is disproportionately attracting a particular age segment. In contrast, customers under 20 represent a much smaller share across all channels, reflecting the business's primary focus on adult consumers.





Promotion and discount analysis indicates that the majority of business performance is generated without promotional incentives. Approximately 26,000 orders were placed without promotions compared with around 7,000 orders with promotions. Similarly, products sold without discounts generated approximately $9.7M in revenue and $3.9M in profit, substantially outperforming discounted sales ($1.0M revenue and $0.4M profit).

This findings suggest that even though the company does not launch promotions or discounts, customer purchasing decisions are not primarily affected.

Promotion Applied	Discount Applied	_Total Revenue	_Total Profit	_Profit Marg
No	No	$7,750,998.91	$3,121,059.01	40.27
No	Yes	$820,685.77	$334,732.17	40.79
Yes	No	$1,926,700.25	$773,737.16	40.16
Yes	Yes	$220,070.67	$89,277.21	40.57
Total		**$10,718,455.60**	**$4,318,805.55**	**40.29**

Profit margins remained stable across all four groups, ranging from 40.16% to 40.57%, indicating that promotions and discounts did not reduce revenue but instead generated positive profits. Moreover, suggesting that more promotion and discount campaigns should be tested and launched to boost purchase power.

The company currently performs strongly without relying heavily on promotional incentives, with the majority of revenue and profit generated from non-promotional sales. However, the stable profit margins across promotional and non-promotional transactions (40.16%–40.57%) suggest that discounts and promotions have not materially eroded profitability. Therefore, the business has an opportunity to strategically expand and test targeted promotional campaigns to stimulate purchase volume and customer engagement while maintaining healthy margins. Future campaigns should be evaluated based on their incremental revenue, conversion, and profitability impact to determine which promotional strategies generate sustainable growth.

Key Insights

1. **Balanced marketing performance:** Email, Referral, Organic, and Ads all contribute comparable revenue and profit, indicating a diversified customer acquisition strategy with no overreliance on a single marketing channel.

2. **Email is the strongest-performing channel:** Email generates the highest revenue (**~$2.6M**) and profit (**~$1.0M**) while recording the lowest churn rate (**61.17%**) among all marketing channels.
3. **Customer retention remains a challenge:** Churn rates remain consistently high across all marketing channels (**61–66%**), suggesting that improving customer retention should be prioritised regardless of acquisition source.
4. **Sales are driven by regular purchases:** More than **26K orders** were completed without promotions, compared with approximately **7K promotional orders**, indicating that most customers purchase without promotional incentives.
5. **Opportunity to optimise promotional strategy:** Products with promotions and discounts create good profit to the company which could be potential opportunities to improve customer purchase power and improve the customer retention. Future analysis should focus on measuring the incremental impact of promotional campaigns and identifying customer segments that respond most effectively to marketing incentives.

5. Recommendations

1) Optimise revenue through seasonal demand planning

The analysis identified recurring demand peaks, particularly in March, May, and November, with November 2010 recording the highest order volume. The company should use these historical patterns to improve seasonal forecasting, inventory planning, staffing, and marketing allocation. Marketing and inventory resources should be increased ahead of anticipated peak periods to capture additional demand while reducing the risk of stock shortages.

Expected impact: Higher peak-season revenue, improved inventory availability, and more efficient resource allocation.

2) Invest in high-performing and high-potential product categories

Toys, Stationery, Clothing, and Home Decor are major contributors to sales volume and revenue and should remain core categories for growth. However, investment should be based on both revenue and profitability, rather than sales volume alone.

In particular:

Stationery represents an attractive growth opportunity due to its combination of strong sales, relatively low production costs, and low returns.

Clothing should be further investigated because it generates revenue comparable to higher-volume categories while carrying relatively high production costs.

Toys should remain a major revenue driver but requires attention to its high return volume.

Expected impact: Increased revenue from high-performing categories while improving category-level profitability and operational efficiency.

3) Develop targeted customer retention strategies

The 63.78% churn rate is the most significant customer-related challenge identified in the analysis. Because churn remains above 60% across customer segments and marketing channels, retention should be addressed across the broader customer base rather than targeting only one demographic.

The company should develop customer segmentation based on purchase frequency, monetary value, recency, age group, and previous purchasing behaviour. This can support targeted strategies such as:

Loyalty and rewards programs for high-value customers

Personalised product recommendations

Re-engagement campaigns for inactive customers

Post-purchase engagement for first-time customers

Targeted incentives for customers showing signs of potential churn

The 60+ segment also represents a potential growth opportunity because it combines meaningful sales volume with relatively low returns.

Expected impact: Lower churn, higher repeat-purchase frequency, and increased customer lifetime value (CLV).

4) Prioritise Email while maintaining a diversified marketing mix

The analysis shows relatively balanced performance across Email, Referral, Organic, and Ads, reducing dependence on a single acquisition channel. However, Email is the strongest-performing channel, generating approximately $2.6M in revenue and $1.0M in profit, while also recording the lowest churn rate.

The company should therefore maintain a diversified acquisition strategy while increasing the strategic use of Email for both acquisition and customer retention. Email campaigns could include personalised recommendations, post-purchase communication, loyalty offers, and win-back campaigns.

Marketing performance should also be evaluated using revenue, profit, acquisition volume, churn, and customer lifetime value, rather than revenue alone.

Expected impact: Improved marketing ROI, stronger customer engagement, and better retention without excessive dependence on one acquisition channel.

5) Use targeted promotions rather than broad discounting

Most orders and revenue are generated without promotional incentives, demonstrating that the business does not currently depend heavily on discounts to generate sales. However, promotional and discounted sales still generate positive revenue and profit, while profit margins remain relatively stable at approximately 40%.

Therefore, the company should not simply increase discounting. Instead, it should test targeted promotional campaigns for specific customer segments, products, or periods.

Expected impact: Increased purchasing activity while protecting profit margins and identifying promotions that generate genuine incremental value.

6) Implement a targeted return-reduction strategy

Returns represent a significant opportunity for operational improvement, with returned products accounting for more than $500K in production costs.

The analysis indicates two different return problems:

Kitchen: Highest return rate despite relatively lower sales volume.

Toys: Highest return volume because of its very high sales volume.

The company should therefore investigate returns at the product and subcategory level to identify specific products driving the problem. Potential causes should be assessed, including product quality, product descriptions, customer expectations, packaging, and delivery-related issues.

Return analysis should focus on:

Return rate × sales volume × production cost

This will help identify products that create the greatest financial impact and allow management to prioritise corrective action.

Expected impact: Lower return rates, reduced production losses, improved customer satisfaction, and stronger profitability.

Recommended strategic priorities			
Priority	**Strategic recommendation**	**Primary objective**	**Expected business outcome**
1	Reduce customer churn through targeted retention strategies	3, 5	Higher CLV and repeat purchases
2	Reduce high-impact product returns	7	Lower costs and protect profitability
3	Scale high-performing categories and optimise category costs	2	Revenue and margin growth
4	Leverage Email for acquisition and retention	4	Higher marketing ROI and engagement
5	Optimise promotions through controlled testing	6	Incremental sales without margin erosion
6	Align inventory and marketing with seasonal demand	1	Higher peak-period revenue

7	Develop high-potential customer segments such as 60+	3	Additional revenue and customer value

6. Interactive Features

A **Year slicer** enables users to examine sales performance across different years, making it easier to identify year-over-year trends and compare key business metrics. A **Marketing Channel slicer** allows users to evaluate the performance of specific acquisition channels, including their impact on revenue, profit, customer churn, and promotional effectiveness. Additionally, an **Age Group slicer** provides the flexibility to analyse customer demographics, purchasing behaviour, and retention patterns across different age segments. These interactive features enable stakeholders to customise the dashboard based on their analytical needs and gain more targeted business insights.

Custom report tooltips were implemented to provide more detailed investigations. Also, the dashboard relies on interactive slicers and cross-filtering between visuals, allowing users to explore relationships between sales performance, customer characteristics, and marketing effectiveness in real time.